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Individual Project 2

Introduction

- **Background:** The Ames Housing dataset, a detailed collection of residential property sales in Ames, Iowa from 2006 to 2010, stands as a significant resource for real estate analysis. It includes 2930 observations with 82 variables, encompassing diverse aspects like physical characteristics, location, and sale conditions. Its richness makes it an ideal tool for predicting house prices and understanding the dynamics of a typical American midwestern town's real estate market.
- **Motivation:** Accurate prediction of house prices is crucial for various stakeholders in the real estate sector. This analysis seeks to explore key factors influencing prices, thereby facilitating better decision-making for buyers, sellers, and industry professionals.

Analysis

- **Data Preprocessing:** Key steps included identifying missing values & combining related features to create comprehensive variables like total square footage, enhancing the dataset's interpretability.
- Variable 'Lot.Frontage' was dropped because it contained too many missing values relative to the other variables.

- Insignificant predictors for 'SalePrice' were also dropped with the exception of Total Basement Square Feet. Examples of dropped predictors include Lot Area, Low Quality Finished Square Feet, & Above Ground Living Area Square Feet.
- Two new variables were created Total Square feet = 1st Floor Square Feet + 2nd Floor Square Feet + Total Basement Square Feet + Garage Area & Total Porch Square Feet = Open Porch Square Feet + Enclosed Porch + Screen Porch
- Exploratory Data Analysis (EDA): Boxplots, histograms, scatter plots, & summary statistics for each variable to identify outliers and distribution patterns.
- Most variables did not show homoscedasticity, therefore a log transformation was applied.
- Focused on the log of the remaining predictors: Masonry Veneer Area, Wood Deck Square Feet, Total Square Feet, & Total Porch Square Feet.

*Note: Log Total Square Feet is the only variable that shows a normal distribution. Also, the log of the predictors Masonry Veneer Area, Wood Deck Square Feet, & Total Porch Square Feet have a notable frequency of zero values. * See pages 1- 2 of the appendix.

- Diagnostic Checks: after fitting the multiple linear regression model for Sale Price on the training dataset, model assumptions were tested through residual analysis, QQ plots, and checks for homoscedasticity.

LogMasVnrArea	LogWoodDeckSF	LogTotalSqFt	LogTotalPorchSF
1.198610	1.091985	1.386943	1.118875

- All predictors have a low VIF indicating minimal multicollinearity

- Influential observations were left in the dataset.

* See pages 3-4 in the appendix

Results

- The core of the analysis involved multiple linear regression (the full model), AIC / BIC models, LASSO model & the Elastic Net for regularization & variable selection. This approach balanced model complexity and predictive accuracy.

Model Insights:

```
Call:
lm(formula = SalePrice ~ ., data = train_set)

Residuals:
    Min       1Q   Median       3Q      Max
-320944  -24970   -3271   19514  386668

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1238819.3    29133.7  -42.522  < 2e-16 ***
LogMasvnrArea    3617.4      414.5    8.727  < 2e-16 ***
LogWoodDecksF    4036.5      398.3   10.134  < 2e-16 ***
LogTotalsqFt    174953.5    3776.3   46.329  < 2e-16 ***
LogTotalPorchSF   2526.1      478.2    5.282  1.4e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 46120 on 2175 degrees of freedom
Multiple R-squared:  0.6564,    Adjusted R-squared:  0.6558
F-statistic: 1039 on 4 and 2175 DF,  p-value: < 2.2e-16
```

- The full model, AIC, & BIC models were more or less the same with similar summaries & coefficient estimates as this one to the left.

```

a0          Length Class      Mode
beta       240    dgCMatrix S4
df          60    -none-    numeric
dim         2     -none-    numeric
lambda     60    -none-    numeric
dev.ratio  60    -none-    numeric
nulldev     1     -none-    numeric
npasses     1     -none-    numeric
jerr        1     -none-    numeric
offset      1     -none-    logical
call        4     -none-    call
nobs        1     -none-    numeric
```

- Lasso & Elastic Net also shared similar summaries as this one to the left.

Performance Metrics:

- Mean Prediction Error: The full model, AIC, & BIC models all had similar mean prediction errors that were lower than the Lasso & Elastic models, and therefore performed better.

Table. Mean Prediction Error by Model

Full Model, AIC, BIC	Lasso	Elastic
2454479690.43122	2608818724.41273	2619573770.18861

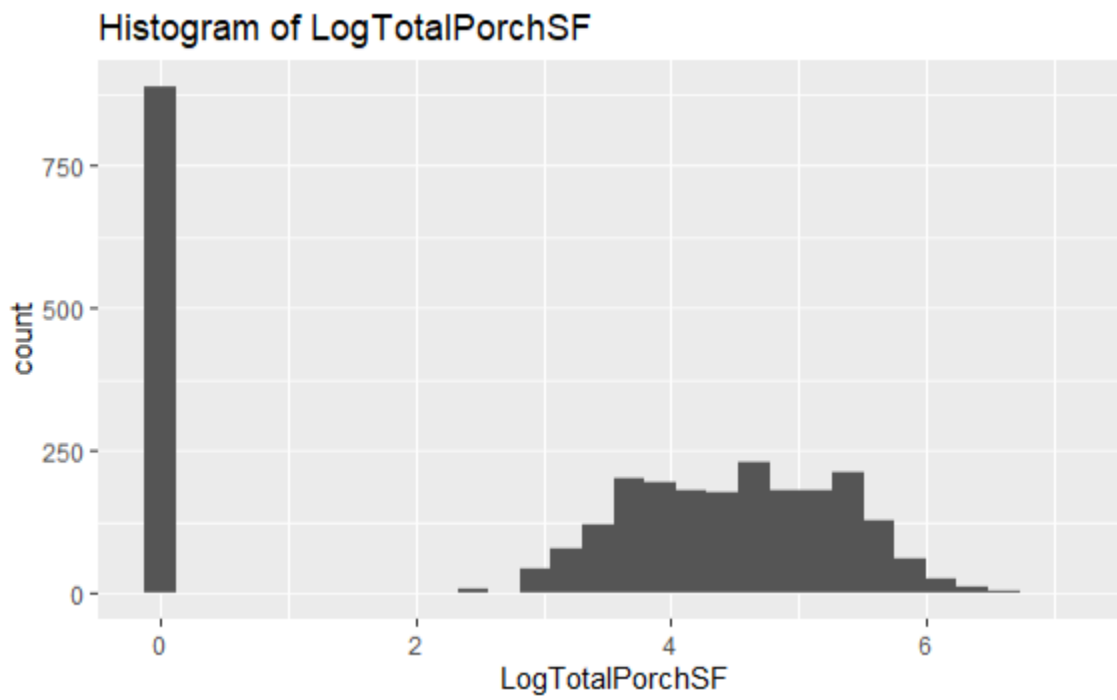
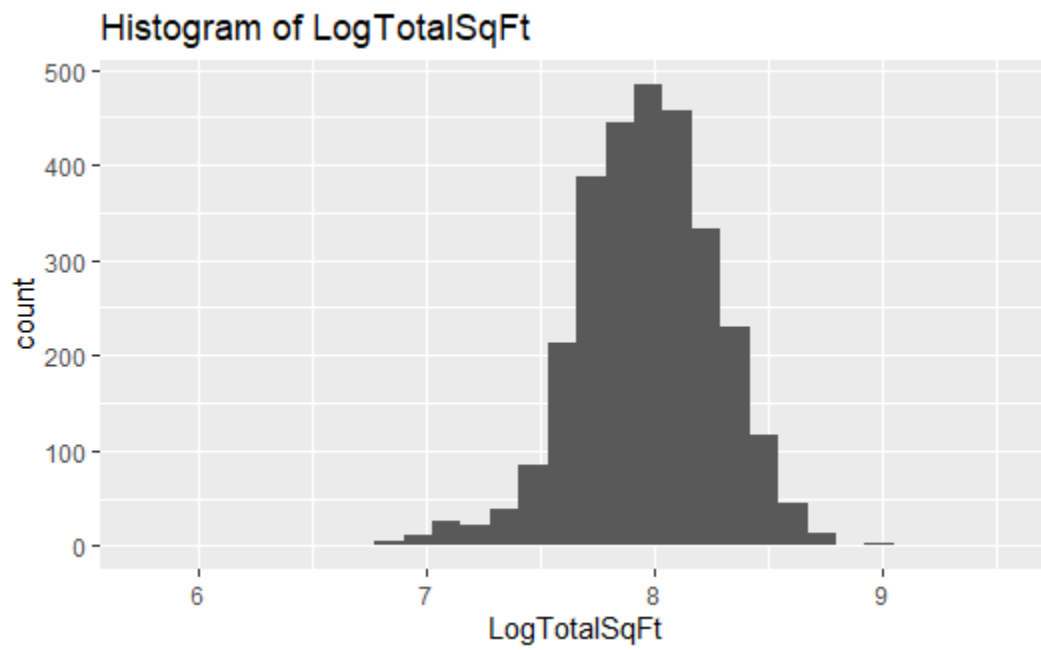
Overall, this analysis & model comparison reveals that Log Masonry Veneer Area, Log Wood Deck Square Feet, & Log Total Square Feet would make for good predictors for Sale Price.

Conclusion

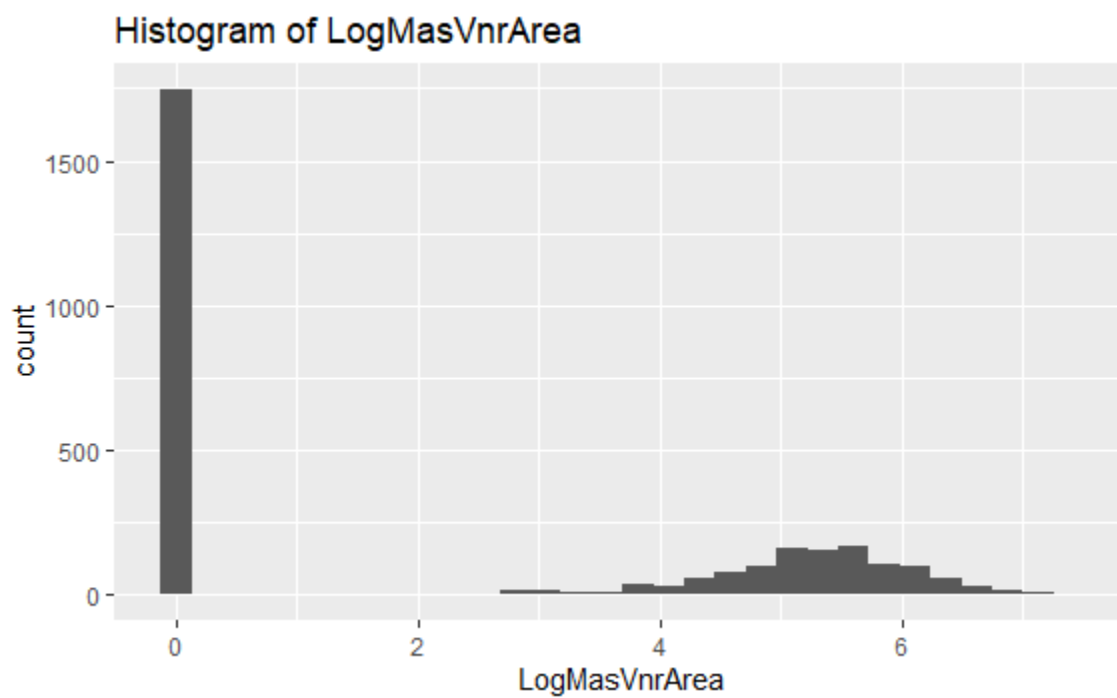
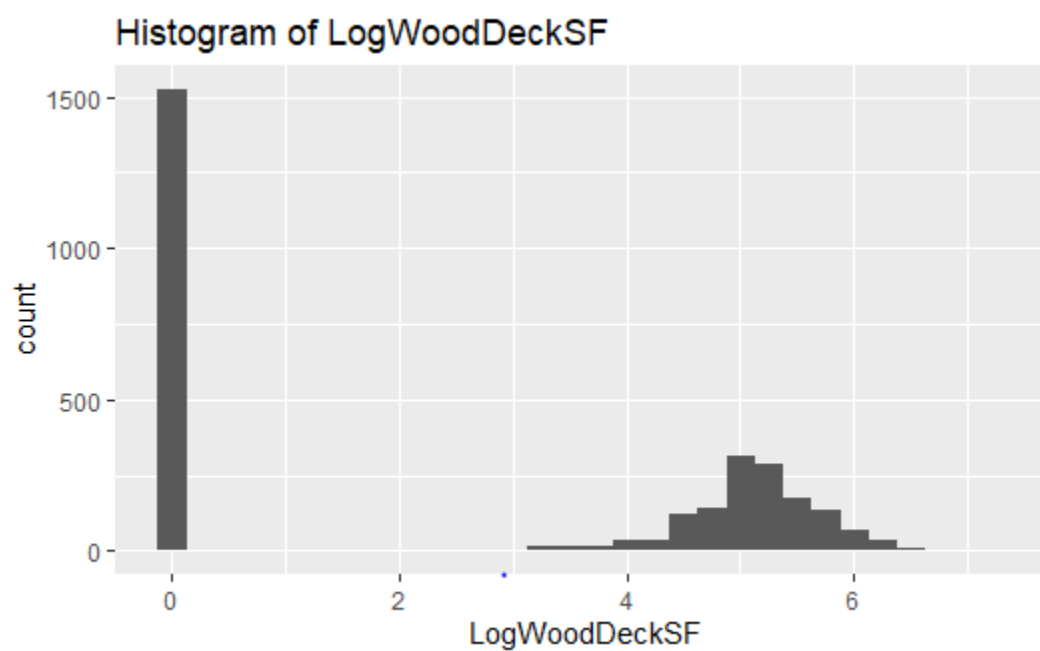
- This analysis provides valuable insights into the Ames housing market, identifying key factors influencing house prices. This understanding can empower stakeholders to make data-driven decisions in real estate transactions.
- Limitations: While these models offer robust predictions, they may not capture all market nuances. Future studies should consider the evolving market trends and potential non-linear relationships.
- Future Work: Incorporating additional data such as economic indicators, neighborhood demographics, and temporal factors could refine predictions. Exploring advanced machine learning techniques might also yield deeper insights.
- These findings underscore the dynamic nature of real estate markets. Continuous analysis and model updating are essential for maintaining relevance in predictive accuracy, making this an ongoing area of research and development.

Appendix

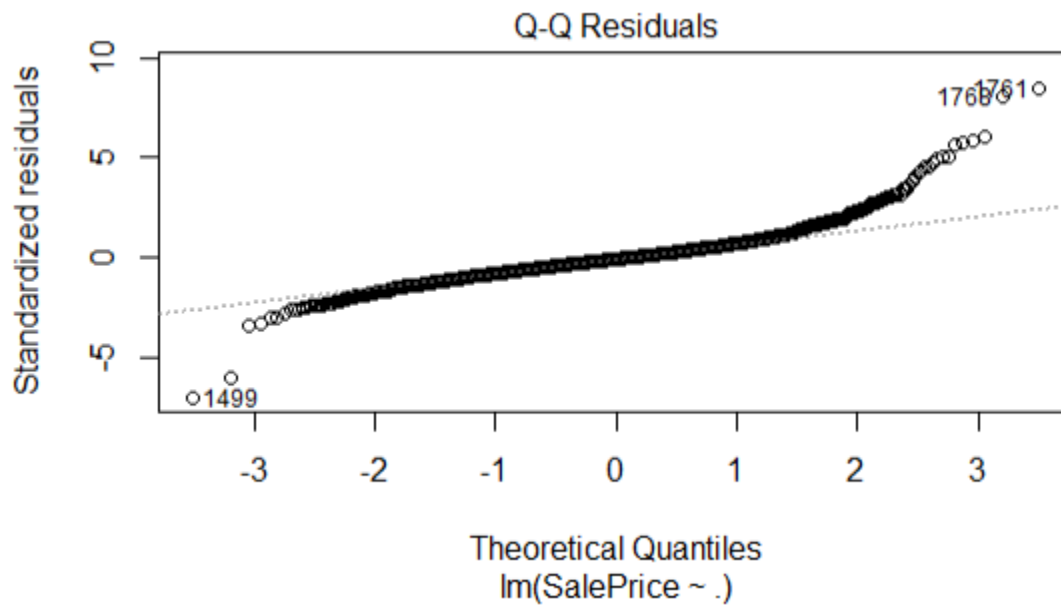
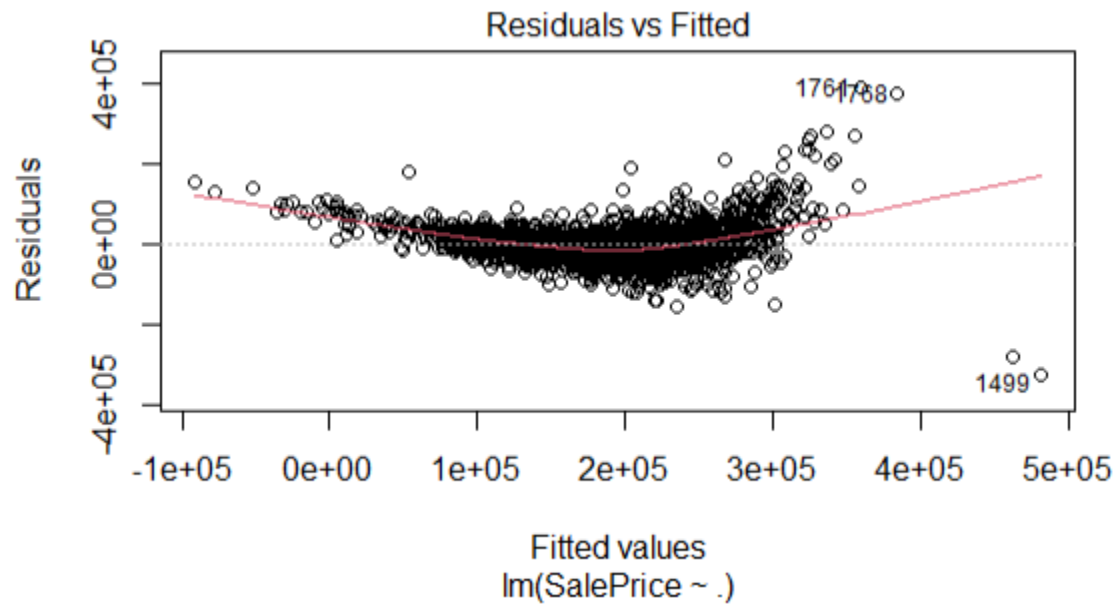
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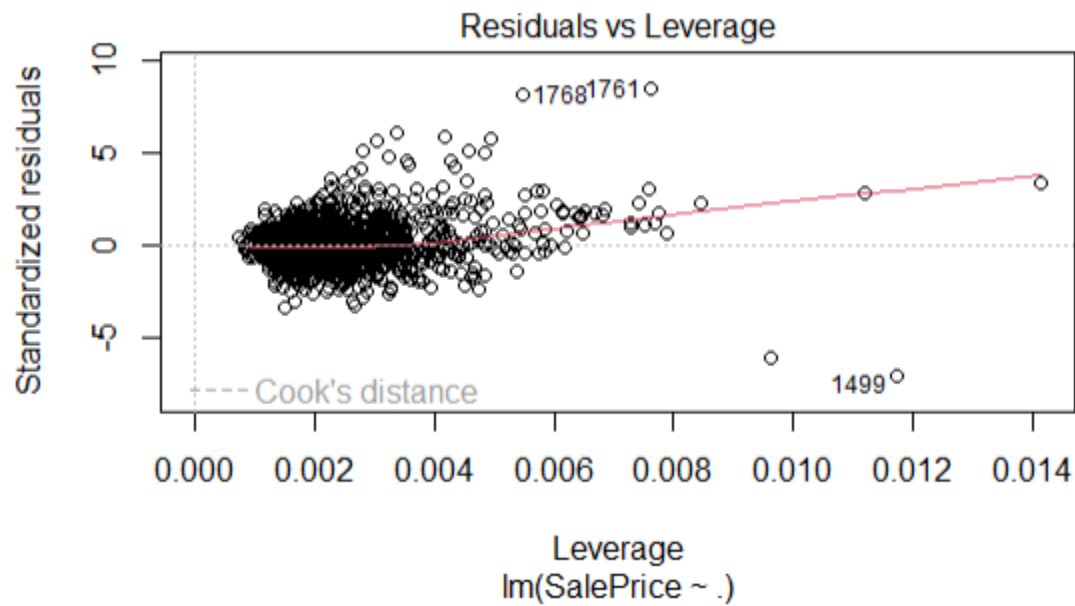
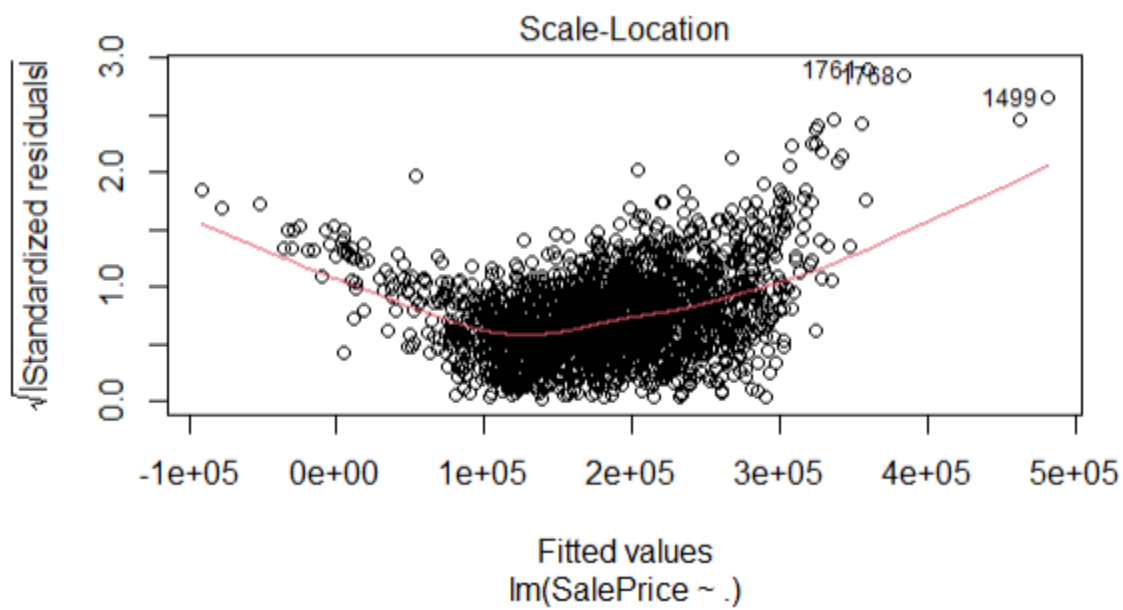
2



3



4



* More in Rmarkdown submission, if necessary.